COURSE OUTLINE
SEMESTER 2
FEBRUARY 2026

LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY
LESOTHO

BIMP2212 – MOBILE DEVICE PROGRAMMING

COURSE DETAILS:

Faculty:	Information and Communication Technology
Programme:	**Degree**: Software Engineering with Multimedia
Year/Semester:	Year 3, Semester 2
Lecturer(s):	Mr Thokoana
Principal Lecturer:	Mr Nthunya
Pre-requisite:	Web Design ~~Principles of Programming Logic and Design~~
Methods of Delivery:	Lectures and tutorial sessions
Credits:	12
Contacts hours:	3 hours (2 lecture +1 tutorial)
Number of weeks:	17 weeks long semester (12 teaching, 1 midterm examination, 1 non-contact 1 revision, 2 final examinations)
No. of assessments	4

Prepared by: Mr. Thokoana
Signature: 28/01/26 Date:

Checked by: Mr. Nthunya
Signature: Date: 28/01/26

This document comprises the following:

- Essential information
- Specific course information
- Course rules & regulations
- Grades
- Plagiarism
- Course introduction
- Course aims & objectives
- Learning outcome
- Specific generic learning skills
- Syllabus + lecture outline
- References
- Assignment schedule
- Assessment criteria
- Specific criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for lecture notes and tutorials
- Submission Requirements + Guideline

1.0 ESSENTIAL INFORMATION

- All courses other than electives are **'significant Courses'**
- As an indicator of workload, one (1) credit carries and additional two (2) hours of self-study per week. For example, a course worth 12 credits require that the student spends an additional 24 hours in a semester through reading, completing the assessment or doing self-directed research for that course.
- Writing and submission of **ALL** assessments work is compulsory in this Course. A student cannot pass this Course without having to submit ALL assignment work by the due date or an approved extension of that date.
- All assignments are to be handed on time on the due date. Students will be penalised 10 percent for the first day and 5 percent per day thereafter for late submission (a weekend or a public holiday count as one day). Late submission, after the date Board of Studies meeting will not be accepted.
- Due dates, compulsory assignment requirements and submission requirements may only be altered with the consent of the majority of students enrolled in this Course at the beginning/early in the program.
- Extensions of time for submission of assignment work may be granted if the application for extension is accompanied by a medical certificate.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- Only the Head of School can grant approval for extension of submission beyond the assignment deadline.
- Re-submission of work can only receive a 50% maximum pass rate.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.
- Harvard referencing and plagiarism policy will apply on all written assignments.

2.0 SPECIFIC COURSE INFORMATION

- Attendance rate of 80% is mandatory for passing Course.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 COURSE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the **Unit Aims and Objectives**
- All assignments must be completed and presented for marking by the due date.
- Marks will be deducted for late work and invalid reasons.
- All assignments must be delivered by the student in person to the lecturer concerned. No other lecturer is allowed to accept students' assignments.
- Assignments may be submitted through the designated learning management system in place.
- All tests/examinations are compulsory.
- Students must sit for the test/examination on the notified date.
- Students are expected to familiarize themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is TOTALLY the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials WITHOUT appointments will also NOT be entertained.
- Note that every assignment is given an ample time frame for completion. This, together with advanced information pertaining deadlines gives you NO EXCUSE not to submit assignments on time.

4.0 GRADES

All Courses and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PX or F:

Grade	Numeric	Grade description
90 – 100	A+	
85 – 89	A	Pass with Distinction
80 – 84	A-	
75 – 79	B+	
74 – 70	B	Pass with Credit
65 – 69	B-	
60 – 64	C+	
55 – 59	C	
50 – 54	C-	Pass
50	PX	Pass after extra work
0 – 49	F	
	EXP	Exempted
	PP	Pass Provisional with extra work needed
	PX	Pass after extra work is given and passed
	X	Ineligible for assessment due to unsatisfactory attendance
	DEF	Deferred
	W	Withdraw
	DNA	Did Not Attend Course
	DNC	Did Not Complete Course

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

5.0.1 Academic Dishonesty

Academic Dishonesty or academic misconduct is any type or cheating that occurs in relation to a formal academic exercise. It can include:

Plagiarism: The adoption or reproduction of original creations of another author (person, collective, organization, community or other type of author, including anonymous authors) without due acknowledgement.

Fabrication: The falsification of data, information or citations in any formal academic exercise.

Deception: Providing false information to an instructor concerning a formal academic exercise – e.g., giving a false excuse for missing a deadline or falsely claiming to have submitted work

Cheating: Any attempt to give obtain assistance in a formal academic exercise (like an examination) without due acknowledgement.

Bribery: or paid services. Giving certain test answers for money.

Sabotage: Acting to prevent others from completing their work. This includes cutting pages out of library books or wilfully disrupting the experiments of others.

Students involved in academic dishonesty will be treated as follows:

1. Plagiarism and fabrication in Assignments, Projects or Presentations: A zero (0) mark will be awarded to the work.

2. **Cheating:** Any form of assistance is a test will be rewarded a zero (0) mark. While in an examination the student automatically repeats the module.
3. **Bribery,** Sabotage or Professional Misconduct: Committing any of these offenses will lead to discontinuity from the University for a certain period depending on the seriousness of the offence.

5.0.2 Copyright, Patents and Ownership of Work

All use of the borrowed work, whether text or images must be referenced and acknowledged. Limkokwing is not liable for any infringement of copyright within students' work. Students are to abide by International Copyright laws. Students engaged in a project activity will be required, as a condition of acceptance as a student, to agree to assign to Limkokwing their right, title and interest in any invention or visual work arising from their studies with the University. Limkokwing has the right to keep students work for use in archives, exhibitions and events. Where students wish to keep original work, they must apply in writing and gain permission from the Academic Management.

6.0 COURSE INTRODUCTION

This course is designed to introduce students to the fundamentals of React Native mobile app development. It is ideal for students who have limited knowledge of programming and no prior experience with JavaScript. The course covers 10 essential topics in React Native, including components, JSX, layout and styling, navigation, handling user input, data management, working with images and videos, debugging, and integrating with native modules.

Throughout the course, students will learn how to build dynamic and interactive mobile apps using React Native, and will gain hands-on experience in debugging and troubleshooting issues. The course will provide students with a strong foundation in React Native, and equip them with the skills and knowledge necessary to build professional mobile applications.

7.0 COURSE AIMS AND OBJECTIVES

1. To provide a comprehensive introduction to JavaScript, including its syntax, data types, and control structures, so that students have a strong foundation for developing React Native apps.
2. To give students an in-depth understanding of cross platform Mobile App Development in React Native, including how to build user interfaces, handle user input, navigate between screens, and fetch data from APIs.
3. To equip students with the skills to store and retrieve data in a mobile app, and to debug and troubleshoot common issues that may arise.
4. To teach students about styling, including the use of stylesheets and gestures, so that they can create visually appealing and interactive apps.
5. To cover advanced topics; including testing, deployment, and working with native modules to build complex, professional-quality apps.

8.0 LEARNING OUTCOME

By the end of this course, students will be able to:

1. To be able to write and understand basic JavaScript code, including syntax, data types, variables, functions, and control structures.
2. To be able to build basic and complex user interfaces in React Native, including handling user input, navigating between screens, and fetching data from APIs.
3. To be able to store and retrieve data in a React Native app, and to debug and troubleshoot common issues that may arise.
4. To be able to style and animate their React Native apps, including the use of stylesheets and gestures, to create visually appealing and interactive apps.

5. To have a solid understanding of advanced topics in React Native, including testing, deployment, and working with native modules, and will be able to build complex, professional-quality apps.

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of the Course, student will acquire skills in:

- Problem solving - Students will learn how to identify and debug common issues in mobile apps, and develop their skills in finding and implementing solutions to problems.
- Design skills - Students will learn how to create visually appealing and interactive user interfaces, and will develop their skills in design and user experience.
- Structured data storage and retrieval- Students will learn how to store and retrieve data locally and will develop their skills in data management and manipulation.
- Self-directed learning - Students will learn how to continue learning on their own, finding resources, seeking help, and experimenting with new ideas.
- Collaboration skills - Students will work in teams to complete projects, and will develop their skills in teamwork, communication, and collaboration.
- Adaptability - Students will learn about new technologies and techniques in React Native, and will develop their ability to adapt to changing technologies and environments.

10.0 PROGRESSION POLICY

Students can only enroll into a higher semester if they have passed all courses in the preceding semester or have failed not more than two courses. A student cannot register for any course at a higher level if they fail its lower-level unit which in essence is a pre-requisite: If student CGPA is less than 2.0 could not proceed to the next semester.

11.0 UNIT SYLLABUS + LECTURE OUTLINE

Week	1
LECTURE	1: Introduction to JavaScript
Lecture Synopsis:	• What is JavaScript? • Basic syntax and data types • Variables, constants, and assignment • Operators and expressions • Control structures (if/else, switch, loops) • Functions and arrow functions • Basic concepts of object-oriented programming in JavaScript
Handout:	Course Outline, Assignment 1

Week	2
LECTURE	2: Introduction to Mobile Development / React Native
Lecture Synopsis:	• Mobile app categories • Native vs cross-platform • What is React Native • React native advantages & use cases • Setting up development environment • Understanding the structure of a React Native project
Handout:	Exercise 1

Week	3
LECTURE	3: Basic Components and Props
Lecture Synopsis:	• Understanding components in React Native • React fundamentals JSX, components, state, props, hooks • Understanding props and their usage • Different types of props • How to pass props between components • Stateless and stateful components • Common components used in React Native (Text, View, Image, etc.)
Handout:	Exercise 2

Week	4
LECTURE	4: Layout
Lecture Synopsis:	• Understanding the Flexbox layout system • Using the style prop to style components • Understanding stylesheet objects and their usage • Creating and using different styles for different screen sizes • Understanding the Dimensions API • Using the PixelRatio API for pixel-perfect styling • Common layout techniques used in React Native
Handout:	Exercise 3
Submission	Assignment 1

Week	5
LECTURE	5: Styling
Lecture Synopsis:	• Introduction to styling in React Native • Understanding color, font size, and other styling options • Understanding the use of themes and color palettes • Using pre-built UI libraries for styling • Best practices for styling in React Native • Creating custom styles and themes • Understanding responsive design in React Native
Handout:	Assignment 2

Week	6
LECTURE	6: User Input and Interaction
Lecture Synopsis:	• Understanding user inputs in React Native • Using buttons and touch events • Implementing basic forms • Working with text inputs and keyboard events • Implementing checkboxes and switches • Creating pickers and modals
Handout:	Group Project, Exercise 4
Submission:	Assignment 2

week 7:	Mid-Term Examinations
week 8:	[semester break]

Week	9
LECTURE	9: Navigation
Lecture Synopsis:	• Introduction to navigation in React Native • Using the Stack, Tabs Navigator for basic navigation • Understanding the lifecycle methods of a screen • Navigating between screens and passing data • Customizing the header and navigation bar • Implementing a drawer navigation
Handout:	Exercise 5

Week	10
LECTURE	10: local storage
Lecture Synopsis:	• Using AsyncStorage for local storage in React Native • Storing and retrieving data with key-value pairs • Encrypting sensitive data in storage
Handout:	Exercise 6

Week	11
LECTURE	11: Working with APIs and Data
Lecture Synopsis:	• Introduction to APIs and data in React Native • Using the fetch API to make network requests • Understanding RESTful APIs • Parsing JSON data • Handling errors and exceptions
Handout:	

Week	12
LECTURE	12: State Management
Lecture Synopsis:	• Different approaches to state management • Understanding state updates • Setting up the context API • Using the context API for global state management
Summation	Group Project

Week	13
LECTURE	13: Deployment and Publishing
Lecture Synopsis:	• Building the app for production • Signing and releasing the app • Publishing the app on the app stores • Updating the app
Handout:	

WEEK	15	REVISION
WEEK	16	FINAL EXAMINATIONS
WEEK	17	FINAL EXAMINATIONS

12.0 REFERENCES

Recommended Books: • Learning React Native: Building Native Mobile Apps with JavaScript, 2nd Edition, Bonnie Eisenman, 2017.
Additional Books: • Getting Started with React Native, Ethan Holmes, Tom Bray, 2015. • Eloquent JavaScript: A Modern Introduction to Programming, 3rd Edition, Marijn Haverbeke, 2018
Reading List: • https://reactnative.dev/docs/getting-started • https://reactnative.dev/docs/components-and-apis

13.0 ASSESSMENT SCHEDULE

Assessment description	Issue Date	Due Date	%
Assignment 1	Week 1	Week 4	15
Lab Test	Week 6	Week 6	20
Assignment 2	Week 3	Week 8	25
Group Project	Week 6	Week 13	40
Total			**100**

2-8/01/25

14.0 ASSESSMENT CRITERIA

You will be graded based on your effort in submitting successfully well worked out assignments, projects, tests, and performance in the examinations.

15.0 LEARNING ACTIVITIES

- In-class coding challenges and exercises to reinforce concepts and practice problem-solving skills
- Group projects to build real-world applications and collaborate with classmates
- Pair programming sessions to work on challenging tasks and learn from each other
- Use of online resources such as tutorials, forums, and documentation to expand knowledge and skills.
- Independent research projects where students delve deeper into specific topics and present their findings to the class

16.0 SPECIFIC CRITERIA

- Each assignment will be handed out with the project brief and will vary, depending on the teaching and learning objectives of the specific assignment.
- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.
- Feedback for each assessment will be issued out within 2 weeks of hand-in date.